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328654(28)

B. E. (Sixth Semester) Examination, Nov.-Dec., 2015

(New Scheme)

(ET & T Engg. Branch)

VLSI DESIGN

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Attempt all questions. Part (a) of each question is compulsory. Attempt any two parts from (b), (c) and (d). Assume suitavble data if required.*

1. (a) ASICs are used for 2

(b) Consider the two FET chain as shown in figure.

Find the output voltage V_{out} for following values

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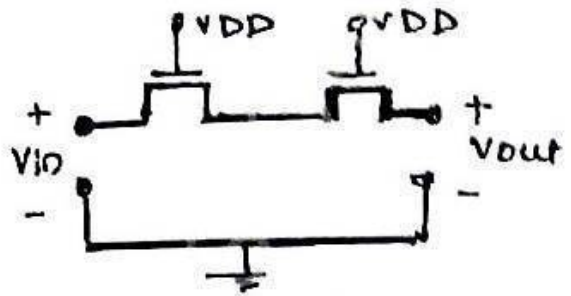
(i) $V_{in} = 3{\cdot}0$ V

(ii) $V_{in} = 1{\cdot}4$ V

Assuming that $V_{DD} = +3{\cdot}3$ V and $V_{th} = 0{\cdot}55$ V.



(c) Design a 4 : 1 multiplexer circuit by using *TG* switches.

(d) Consider a CMOS process with the following parameters

$K_n' = 140\ \mu A/V^2$ $Kp^1 = 60\ \mu A/V^2$

$V_{Tn} = +0{\cdot}7$ V $V_{TP} = -0{\cdot}7$ V and $V_{DD} = 3$ V

Verify the dependence of VM on device ratio.

2. (a) Where the beading effect is caused?

(b) Define the process to find Eulerian path with suitable example.

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(c) Draw the physical structure of NOT gate considering horizontal FETS only. 7

(d) Write a short note on active areas and doped silicon regions. 7

3. (a) How do we check the compatibility between schematic and layout by using EDA tool? 2

(b) Draw the transmission gate-based CMOS 4 way Data selector. (Stick diagram and truth table only). 7

(c) Explain the operation of positive edge triggerd DFF with suitable schematic diagram and wave forms. 7

(d) Explain operation of 6-T DRAM with suitable schematic diagram. 7

4. (a) In a CPLD X C 9572 – 7 PC 84; what do you mean by 72? 2

(b) Write a VHDL to implement a multiplexer circuit (4 : 1) using conditional assignment statement in data flow style of modeling. Assuming that enable input used in asserted low. 7

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(c) Implement a 4 bit adder in VHDL using for _ generate statement. Also write a VHDL code for the component used.

(d) Write a VHDL code to implement a BCD to 7 segment decodes using common anode 7 segment display.

5. (a) Which state machine is faster?

(b) Demonstrate how T-flip-flop can be used to design counter. Write a structural code to implement 3 bit counter in VHDL using TFF.

(c) Write a test-bench to test a fuel adder circuit.

(d) Write a VHDL code to synthesize a sequence detector for a sequence "110". Write a structural code for the same. Use minimum hardware technique, assuming non-overlaping sequence.

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